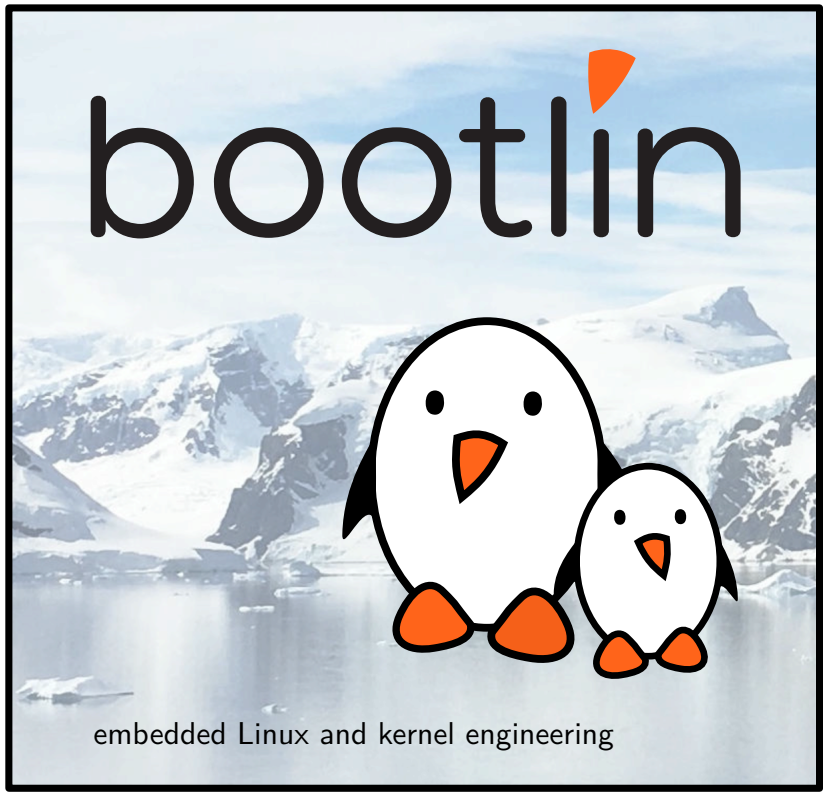
Layers



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Corrections, suggestions, contributions and translations are welcome!



Introduction to layers

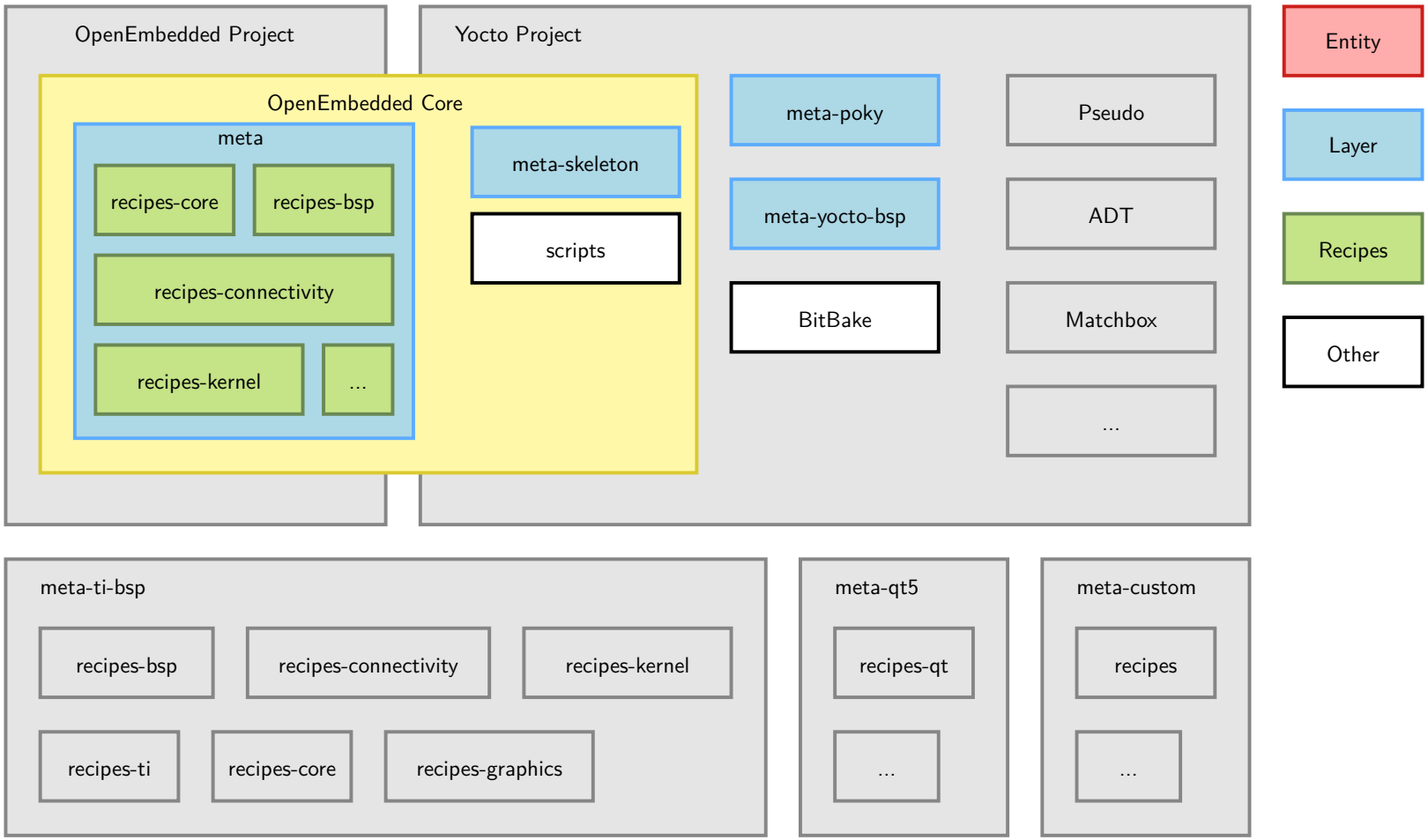


Layers principles

- The OpenEmbedded *build system* manipulates *metadata*.
- Layers allow to isolate and organize the metadata.
  - A layer is a collection of recipes, classes, configuration, etc.
- It is a good practice to begin a layer name with the prefix `meta-`.



Official Yocto layers



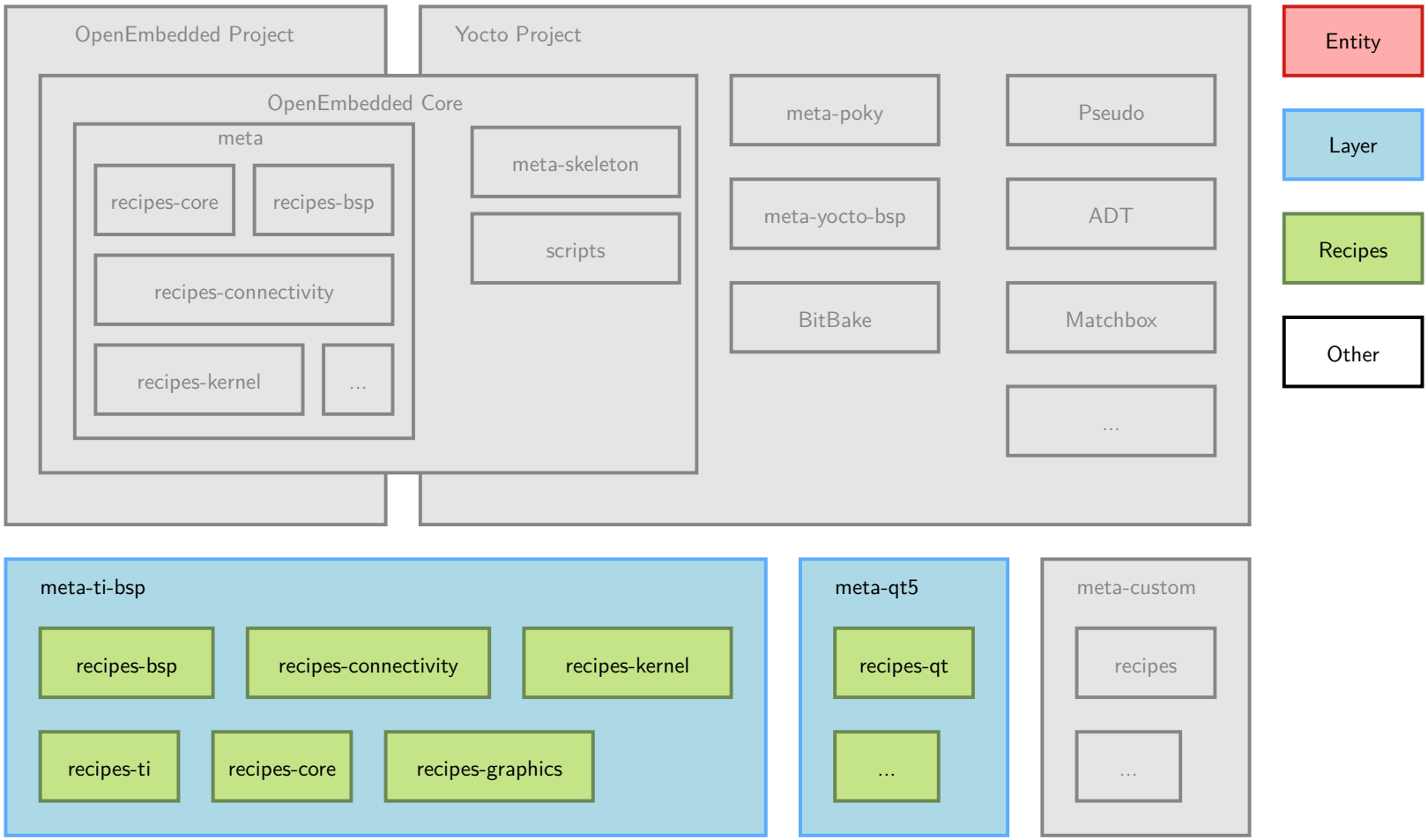


Official Yocto layers

- The Yocto project provides a set of basic common layers:
  - meta
  - meta-skeleton
  - meta-poky
  - meta-yocto-bsp
- This is not a final set of layers. It is the common base.
- Layers are added when needed.
- When making modifications to the existing recipes or when adding new ones, it is a good practice to avoid modifying the layers. Instead you can create your own ones!



Third party layers





Integrate and use a layer 1/3

- A list of existing and maintained layers can be found at https://layers.openembedded.org
- Instead of redeveloping layers, always check the work hasn't been done by others.
- It takes less time to download a layer providing a package you need and to add an append file if some modifications are needed than to do it from scratch.



Integrate and use a layer 2/3

- The location where a layer is saved on the disk doesn't matter.
  - But a good practice is to save it where all others layers are stored.
- The only requirement is to let BitBake know about the new layer:
  - The list of layers BitBake uses is defined in `$BUILDDIR/conf/bblayers.conf`
  - To include a new layer, add its absolute path to the BBLAYERS variable.
  - BitBake parses each layer specified in BBLAYERS and adds the recipes, configurations files and classes it contains.



Integrate and use a layer 3/3

- The `bitbake-layers` tool is provided alongside `bitbake`.
- It can be used to inspect the layers and to manage `$BUILDDIR/conf/bblayers.conf`:
  - `bitbake-layers show-layers`
  - `bitbake-layers add-layer meta-custom`
  - `bitbake-layers remove-layer meta-qt5`



Some useful layers

- Many SoC specific layers are available, providing support for the boards using these SoCs. Some examples: `meta-ti-bsp`, `meta-freescale` and `meta-st-stm32mp`.
- Other layers offer to support applications not available in OpenEmbedded-Core:
  - `meta-firefox`: Firefox web browser support.
  - `meta-filesystems`: support for additional filesystems.
  - `meta-java`: Java support.
  - `meta-arm-toolchain`: Arm GCC toolchain recipes.
  - `meta-qt6`: QT6 modules.
  - `meta-realtime`: real time tools and test programs.
  - `meta-virtualization` and many more...

  Notice that some of these layers do not come with all the Yocto branches. `meta-realtime` layer does not have a `honister` (3.4) branch, for example.



Layer recommendations



Layer recommendations

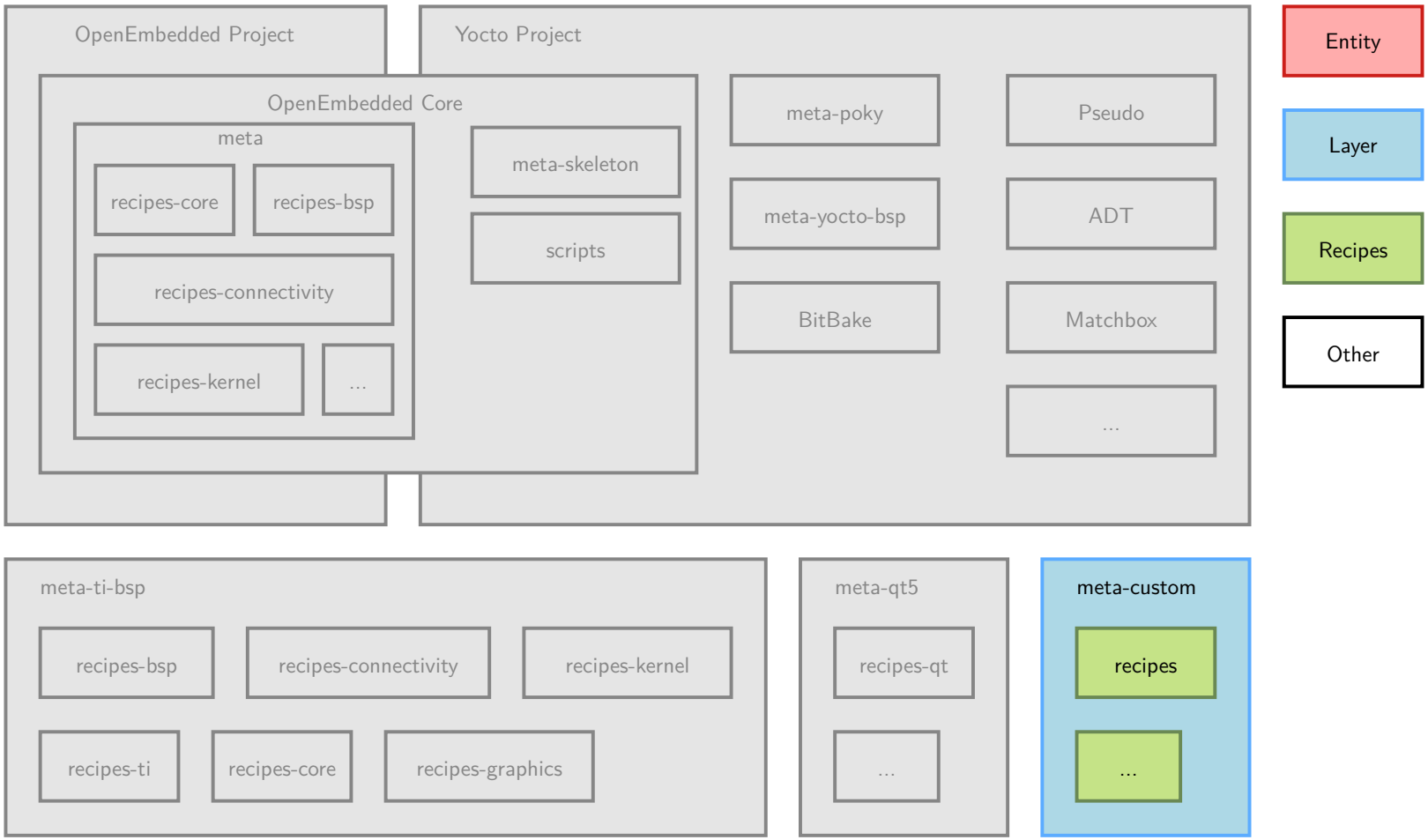
- Keep your build system simple: use very few layers initially
- Then add layers when needed based on benefit/cost ratio
  - The quality of several board/SoM layers is questionable
  - The quality of SoC vendor layers is varying
- A working example:
  - https://github.com/bootlin/yocto-kiss
  - Minimal dependencies: BitBake, OE-core, meta-arm
  - One company-specific layer: meta-kiss
  - Vendor-specific layers: meta-kiss-amd, meta-kiss-nxp, meta-kiss-st, meta-kiss-ti
  - Custom distro and machine configurations, image recipes
  - Can be used as a starting point for your project
  - Introduced at Yocto Project Summit 2023 (slides, video)



Creating a layer



Custom layer





Create a custom layer 1/2

- A layer is a set of files and directories and can be created by hand.
- However, the `bitbake-layers create-layer` command helps us create new layers and ensures this is done right.
- `bitbake-layers create-layer -p <PRIORITY> <layer>`
- The **priority** is used to select which recipe to use when multiple layers contains the same recipe
- The layer priority takes precedence over the recipe version number ordering. This allows to downgrade a recipe in a layer.



Create a custom layer 2/2

- The layer created will be pre-filled with the following files:

  conf/layer.conf

  The layer's configuration. Holds its priority and generic information. No need to modify it in many cases.

  - Mandatory, this is the entry point for the layer.

  COPYING.MIT

  The license under which a layer is released. By default MIT.

  README

  A basic description of the layer. Contains a contact e-mail to update.

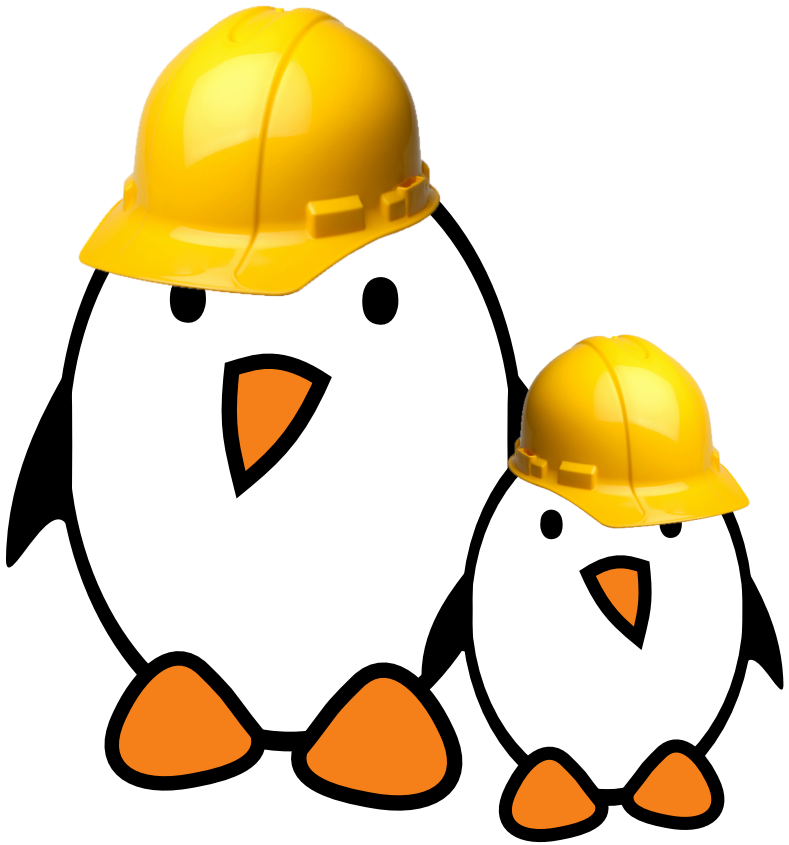
- By default, all metadata matching `./recipes-*/*/*.bb` will be parsed by the BitBake *build engine*.



Creating a layer: best practices

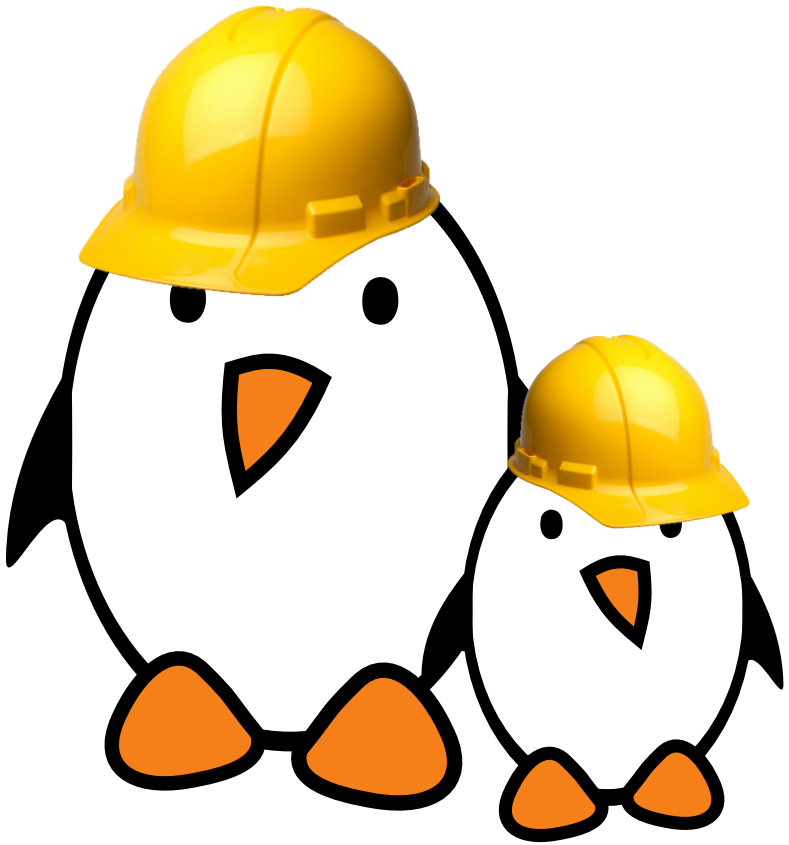
- Do not copy and modify existing recipes from other layers. Instead use append files.
- Avoid duplicating files. Use append files or explicitly use a path relative to other layers.
- Save the layer alongside other layers.
- Use LAYERDEPENDS to explicitly define layer dependencies.
- Use LAYERSERIES_COMPAT to define the Yocto version(s) with which the layer is compatible.

Practical lab - Create a custom layer



- Create a layer from scratch
- Add recipes to the new layer
- Integrate it to the build

Practical lab - Extend a recipe



- Apply patches to an existing recipe
- Use a custom configuration file for an existing recipe
- Extend a recipe to fit your needs